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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Control System I Laboratory

Final Project Report

Section: A1 Group: 05

Dual-Axis Solar Tracking System with Maximum Power Point Tracking (MPPT)

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Table of Contents

1 Abstract 1

2 Introduction 1

3 Design 1

3.1 Problem Formulation 2

3.1.1 Identification of Scope 2

3.1.2 Literature Review 2

3.1.3 Formulation of Problem 2

3.1.4 Analysis 2

3.2 Design Method 3

3.3 Circuit Diagram 3

3.4 Simulation Model 4

3,5 Hardware Design 4

3.6 Full Source Code of Firmware 5

4 Implementation 8

4.1 Description 8

5 Design Analysis and Evaluation 9

5.1 Novelty 9

5.2 Design Considerations 9

5.2.1 Considerations to public health and safety 9

5.2.2 Considerations to environment 9

5.2.3 Considerations to cultural and societal needs 10

5.3 Investigations 10

5.3.1 Data Collection 10

5.3.2 Results and Analysis 11

5.4 Limitations of Tools 11

5.5 Impact Assessment 11

5.5.1 Assessment of Societal and Cultural Issues 11

5.5.2 Assessment of Health and Safety Issues 12

5.5.3 Assessment of Legal Issues 12

5.6 Sustainability Evaluation 12

5.7 Ethical Issues 12

6 Reflection on Individual and Team work 13

6.1 Individual Contribution of Each Member 13

6.2 Log Book of Project Implementation 13

7 Communication to External Stakeholders ……..……...14
7.1 User Manual 14
7.2 Github Link 15
7.3 YouTube Link 15
8 Future Work ……15
9 References……16

1 Abstract

This project presents the design and implementation of a dual-axis solar tracking system integrated with Maximum Power Point Tracking (MPPT) to improve solar energy efficiency. Unlike fixed solar panels, the proposed system continuously aligns the panel with the sun's position along both horizontal and vertical axes, maximizing solar irradiance capture. Additionally, an MPPT algorithm ensures optimal power extraction under varying environmental conditions. The system is implemented using sensors, control circuitry, and embedded programming. Experimental results demonstrate improved efficiency compared to static systems, making it suitable for renewable energy applications.

2 Introduction

The increasing demand for energy and the depletion of fossil fuels have made renewable energy sources, particularly solar energy, an essential alternative. Solar power is clean, abundant, and sustainable; however, its efficiency is significantly affected by environmental conditions and the orientation of the solar panel.

In conventional systems, solar panels are fixed at a specific angle, which limits their ability to capture maximum sunlight throughout the day. As the sun moves across the sky, the angle of incidence changes, resulting in reduced power output. To overcome this limitation, solar tracking systems are used to continuously adjust the panel's orientation to follow the sun.

A dual-axis solar tracking system improves efficiency by allowing movement along both horizontal (azimuth) and vertical (elevation) axes, ensuring that the panel remains perpendicular to sunlight at all times. However, even with optimal positioning, the electrical output of a solar panel varies with changes in temperature, irradiance, and load conditions.

To address this issue, Maximum Power Point Tracking (MPPT) is implemented. MPPT is a technique used to extract the maximum possible power from a solar panel by dynamically adjusting its operating point. By combining dual-axis tracking with MPPT, both mechanical and electrical efficiencies are enhanced.

In this project, a dual-axis solar tracking system integrated with MPPT is designed and implemented using sensors, control circuitry, and embedded programming. The system aims to maximize energy harvesting while maintaining a cost-effective and practical design suitable for real-world applications.

3 Design

The design of the proposed **Dual-Axis Solar Tracking System with MPPT** integrates both mechanical and electrical subsystems to maximize solar energy harvesting. The system is divided into sensing, control, actuation, and power optimization units. The overall objective is to ensure that the solar panel continuously aligns with the sun while operating at its maximum power point.

3.1 Problem Formulation

3.1.1 Identification of Scope

The scope of this project includes the development of a compact and efficient solar tracking system capable of:

- Tracking sunlight in two axes (horizontal and vertical)
- Detecting light intensity using LDR sensors
- Controlling movement using servo motors
- Measuring electrical parameters (voltage and current)
- Implementing MPPT algorithm for maximum power extraction
- Displaying real-time data using an LCD

The system is designed to be low-cost, energy-efficient, and suitable for small-scale applications.

3.1.2 Literature Review

Traditional solar systems are mostly fixed-position panels, which suffer from low efficiency due to constant changes in the sun's position. To improve performance, researchers have developed:

- Single-axis tracking systems, which follow the sun in one direction but still miss optimal alignment
- Dual-axis tracking systems, which provide better alignment but may lack electrical optimization
- MPPT-based systems, which optimize electrical output but do not address physical orientation

This project combines both dual-axis tracking and MPPT, offering a more efficient hybrid solution compared to existing systems.

3.1.3 Formulation of Problem

The main challenges addressed in this project are:

- Loss of solar energy due to improper panel orientation
- Inefficient power extraction due to mismatch between load and panel characteristics
- Variation in output caused by changing environmental conditions

To solve these issues:

- LDR sensors are used to detect the direction of maximum sunlight
- Servo motors adjust the panel orientation accordingly
- MPPT ensures the system operates at the maximum power point

3.1.4 Analysis

The output power of a solar panel depends on both irradiance and angle of incidence. Maximum power is achieved when sunlight strikes the panel perpendicularly.

observations:

- Sun position changes continuously → requires real-time tracking
- Power-voltage characteristics are nonlinear
- Maximum power point varies with temperature and sunlight

Therefore, combining mechanical tracking and electrical optimization (MPPT) significantly improves overall efficiency.

3.2 Design Method

The design methodology follows a systematic approach:

1. Light Detection:
   Four LDR sensors are placed in a cross configuration to detect sunlight intensity differences.
2. Signal Processing:
   The ESP32 reads analog values from the LDRs and determines the direction of maximum light.
3. Motor Control:
   Based on sensor comparison:
   - Horizontal servo adjusts left-right movement
   - Vertical servo adjusts up-down movement
4. Power Measurement:
   Voltage and current sensors continuously monitor solar output.
5. MPPT Implementation:
   A control algorithm (e.g., Perturb & Observe) adjusts the duty cycle of the buck converter.
6. Output Display:
   LCD shows voltage, current, and calculated power.

3.3 Circuit Diagram

The circuit consists of the following interconnected modules:

- **ESP32 Microcontroller** (central control unit)
- **LDR Sensor Network** (sunlight detection)
- **Servo Motors** (dual-axis movement)
- **Voltage & Current Sensors** (data acquisition)
- **LM2596 Buck Converter** (MPPT control)
- **LCD Display (I2C)** (user interface)

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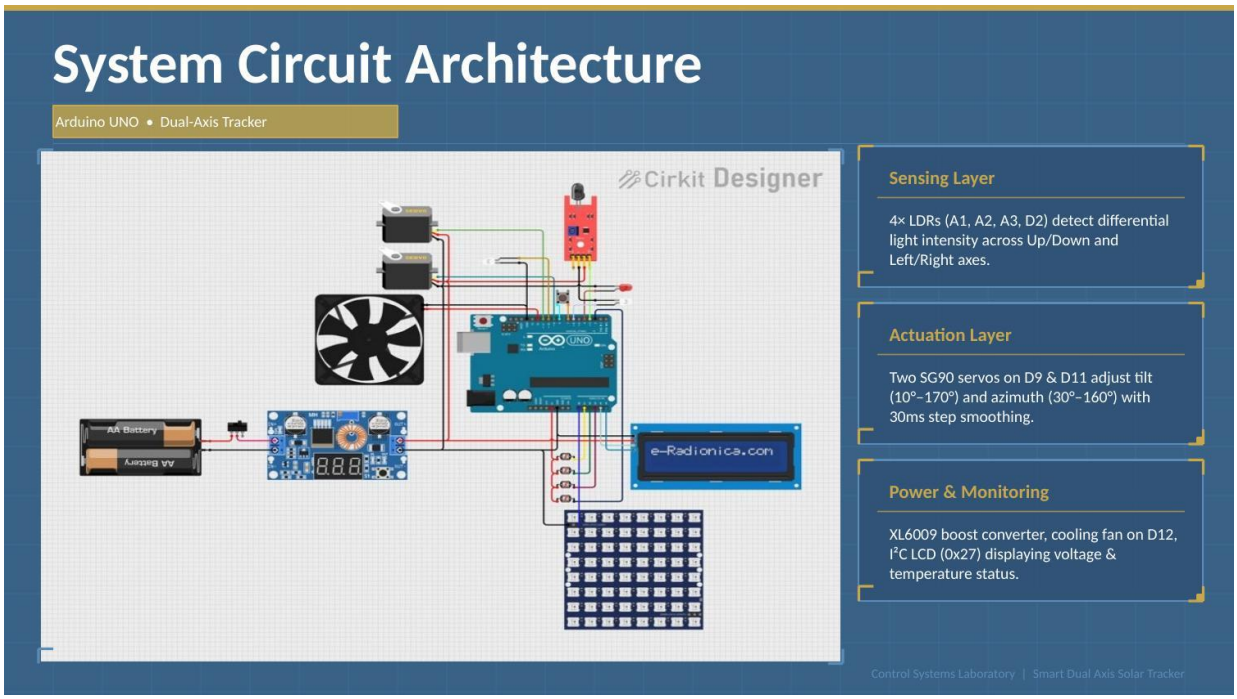
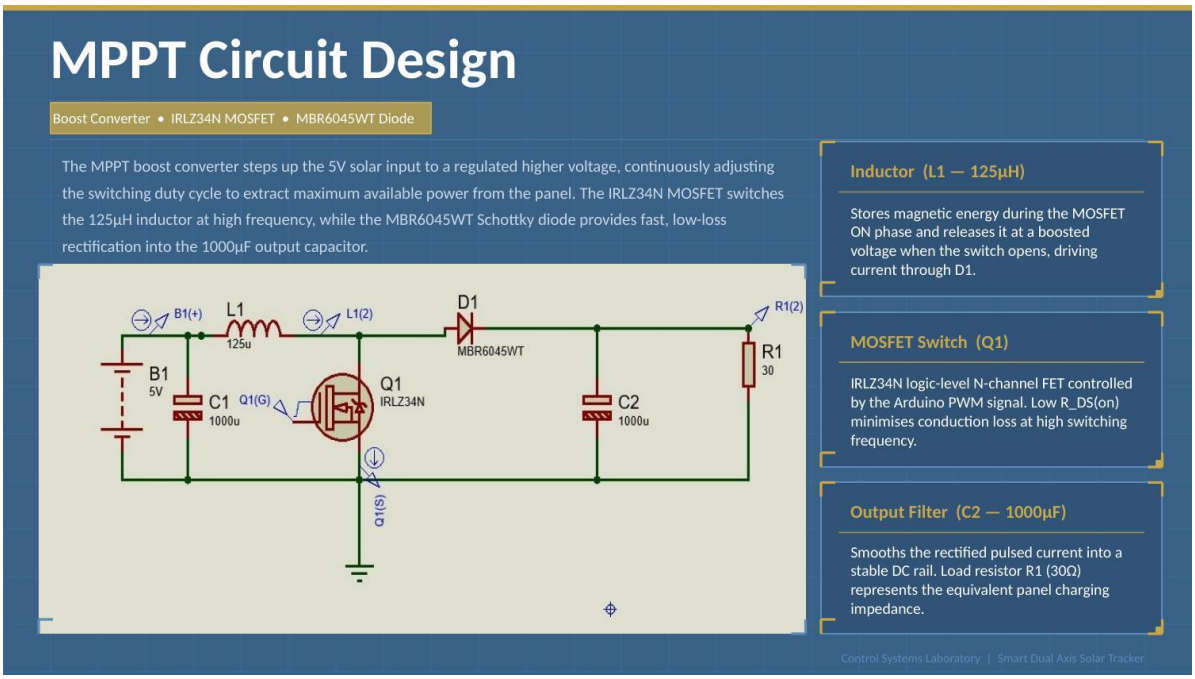
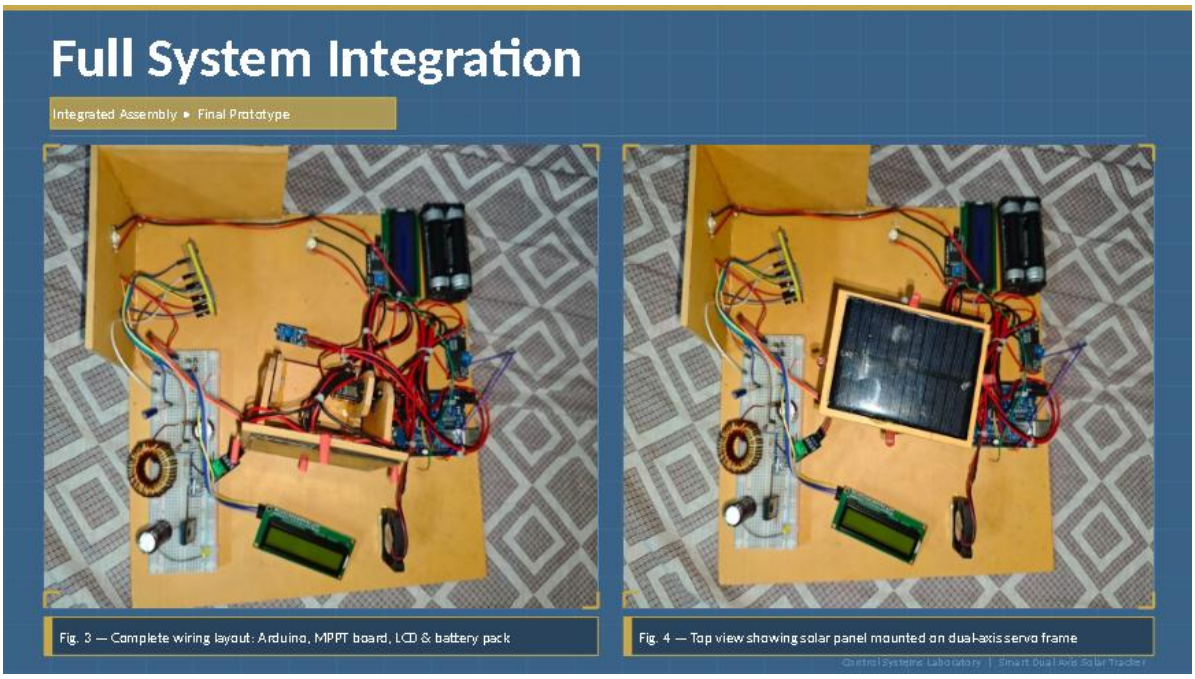


3.4 Simulation Model



3.5 Hardware design



The mechanical design ensures proper movement and stability of the solar panel:

- A dual-axis frame structure allows rotation in both directions
- Servo motors are mounted at appropriate joints

- The solar panel is fixed securely on the top platform
- Sensors are positioned to avoid shadow interference

The design is optimized to minimize vibration and ensure smooth tracking.

3.6 Full Source Code of Firmware

Solar Tracker:

1.Solar Tracking Logic:

```
void updateSensorsAndTargets() {
  LDR_Up_Value    = readLDR(LDR_Up_Pin, LDR_Up_Dig);
  LDR_Down_Value  = readLDR(LDR_Down_Pin, LDR_Down_Dig);
  LDR_Left_Value  = analogRead(LDR_Left_Pin);
  LDR_Right_Value = analogRead(LDR_Right_Pin);

  int diffUD = LDR_Up_Value - LDR_Down_Value;
  int diffLR = LDR_Left_Value - LDR_Right_Value;

  const int deadzone = 100;

  if (abs(diffUD) > abs(diffLR)) {
    if (abs(diffUD) > deadzone)
      targetUpDown += constrain(diffUD / 600, -2, 2);
  } else {
    if (abs(diffLR) > deadzone)
      targetLeftRight -= constrain(diffLR / 600, -2, 2);
  }
}
```

2. Smooth Servo Movement:

```
void servoStepper() {
  if (posUpDown < targetUpDown) posUpDown++;
  else if (posUpDown > targetUpDown) posUpDown--;

  if (posLeftRight < targetLeftRight) posLeftRight++;
  else if (posLeftRight > targetLeftRight) posLeftRight--;

  servoUpDown.write(posUpDown);
  servoLeftRight.write(posLeftRight);
}
```

3. Solar Voltage Measurement + Day/Night Detection:

```
int rawSolar = analogRead(solarPin);
solarBuf[solarBufIdx++] = rawSolar;

long sum = 0;
for (int i = 0; i < ct; ++i) sum += solarBuf[i];
SolarValue = sum / ct;

if (SolarValue > 200) {
  digitalWrite(solarLedPin, HIGH);
  digitalWrite(dayNightLedPin, LOW);
} else {
  digitalWrite(solarLedPin, LOW);
  digitalWrite(dayNightLedPin, HIGH);
}
```

4. Temperature-Based Fan Control:

```
void tempAndFanControl() {
  if (digitalRead(tempSensorPin) == LOW)
    digitalWrite(fanPin, HIGH);
  else
    digitalWrite(fanPin, LOW);
}
```

MPPT (perturb and Observe Method):

1. ADC Data Acquisition & Scaling:

```
void ADCtask(void *pvParameters)
{
    int v_raw  = adc_read_avg(adc1_handle, ADC_CHANNEL_0,
NUM_SAMPLES, 5);
    int i_raw  = adc_read_avg(adc1_handle, ADC_CHANNEL_3,
NUM_SAMPLES, 5);

    mppt_data.mppt_voltage = (int)(calibrated.mppt_voltage /
RATIO);
    mppt_data.mppt_current = (int)((calibrated.mppt_current
- CURRENT_OFFSET_MV)
                                   * 1000 /
CURRENT_SENS_MV_A);

    xQueueOverwrite(mppt_queue, &mppt_data);
}
```

2. MPPT Algorithm (Perturb & Observe):

```
float power_mW = (V_mV * I_mA) / 1000.0f;
float delta_P = power_mW - prev_power_mW;

if (fabsf(delta_P) > 1.0f) {
    if (delta_P > 0.0f) {
        buck_duty += direction;
    } else {
        direction = -direction;
        buck_duty += direction;
    }
}
```

3. PWM Duty Cycle Update (Boost Converter Control):

```
if (buck_duty > BUCK_DUTY_MAX_TICKS) buck_duty =
BUCK_DUTY_MAX_TICKS;
if (buck_duty < BUCK_DUTY_MIN_TICKS) buck_duty =
BUCK_DUTY_MIN_TICKS;

ESP_ERROR_CHECK(
    mcpwm_comparator_set_compare_value(buck_comparator,
buck_duty)
);
```

4. LCD Display Output:

```
snprintf(line1, sizeof(line1), "Pin:%8.1f mW ",
disp.power_in_mW);
snprintf(line2, sizeof(line2), "Duty:      %3d%% ",
disp.buck_duty_pct);

lcd_set_cursor(0, 0);
lcd_print(line1);
lcd_set_cursor(0, 1);
lcd_print(line2);
```

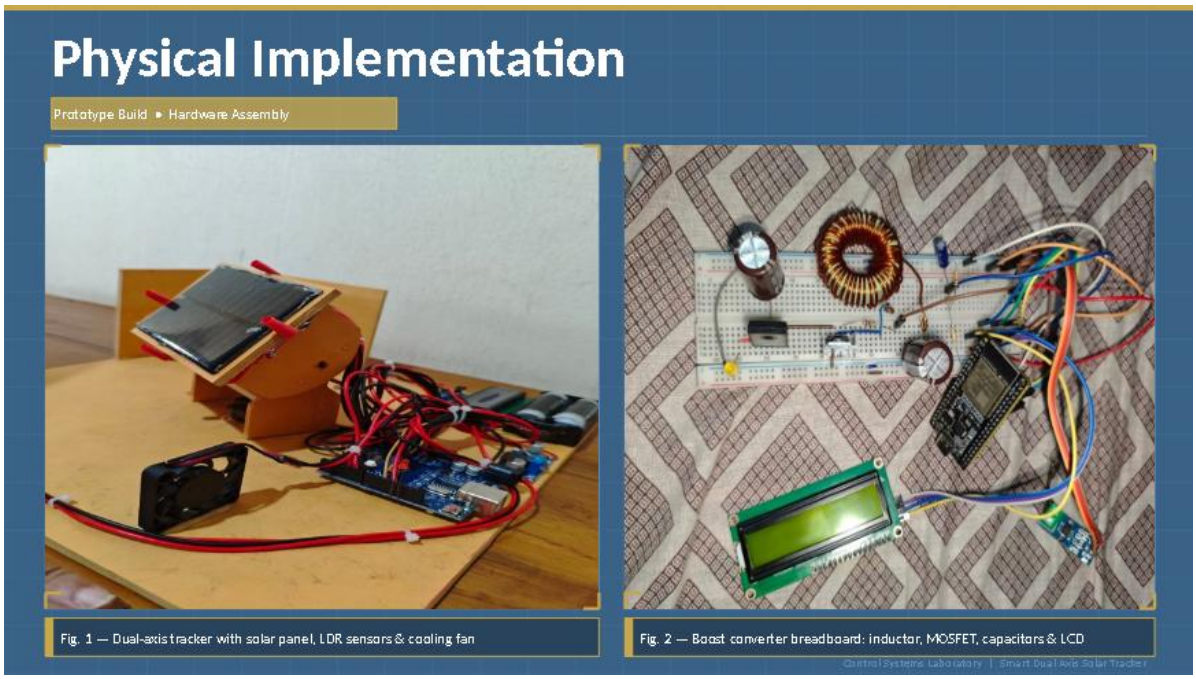
4 Implementation

4.1 Description

Built prototype using:

- Solar panel
- Arduino/controller
- LDR sensors
- DC motors

The implemented system operates by continuously tracking the position of the sun and adjusting the solar panel orientation to maximize energy absorption. At the beginning of the day, when the sun rises in the east, the system automatically rotates the panel towards the east to capture the initial sunlight. As the sun moves across the sky, the system dynamically adjusts the panel's position along both horizontal and vertical axes to maintain optimal alignment.
Around midday, when the sun is nearly overhead, the panel is oriented upwards to receive maximum direct irradiance. As the day progresses towards evening, the system gradually shifts the panel towards the west, following the sun's movement until sunset.
This continuous adjustment is achieved using LDR sensors that detect variations in light intensity from different directions. The controller processes these signals and drives the servo motors accordingly to minimize the difference in light intensity between sensors, ensuring accurate tracking.
Simultaneously, the electrical performance of the solar panel is monitored using voltage and current sensors. The MPPT algorithm adjusts the operating point of the system to ensure maximum power extraction under varying sunlight conditions.
Overall, the system demonstrates real-time solar tracking combined with efficient power optimization, resulting in improved energy output compared to fixed solar panels.



5 Design Analysis and Evaluation

5.1 Novelty

The novelty of this project lies in the effective integration of both mechanical solar tracking and electrical power optimization within a single, low-cost system. While many conventional systems implement either tracking or MPPT independently, this project combines a dual-axis solar tracking mechanism with a Maximum Power Point Tracking (MPPT) algorithm to significantly enhance overall efficiency.

Unlike fixed or single-axis systems, the proposed design continuously aligns the solar panel with the sun in both horizontal and vertical directions, ensuring maximum solar irradiance throughout the day. At the same time, the MPPT technique dynamically adjusts the operating point of the solar panel to extract the highest possible power under varying environmental conditions such as changes in sunlight intensity and temperature.

Another key novelty is the use of a cost-effective and compact implementation using an ESP32 microcontroller, LDR-based sensing, and simple servo motors, making the system suitable for small-scale and educational applications. The integration of real-time monitoring through an LCD display further enhances usability and system transparency.

Overall, the project presents a practical and efficient solution that improves energy harvesting by combining two optimization techniques—mechanical tracking and electrical control—into a single unified system.

5.2 Design Considerations

During the design and implementation of the system, several important factors were considered to ensure safe, sustainable, and socially acceptable operation.

5.2.1 Considerations to public health and safety

The system is designed with safety as a priority:

- **Low-voltage operation** reduces the risk of electrical hazards
- Proper insulation and secure wiring prevent short circuits
- Moving parts (servo motors) are operated at safe speeds to avoid mechanical injury
- Components are mounted securely to ensure structural stability
- Heat generation is minimal, reducing the risk of overheating

These measures ensure that the system can be safely used in residential or educational environments.

5.2.2 Considerations to environment

The project promotes environmental sustainability in several ways:

- Utilizes solar energy, a clean and renewable source of power
- Reduces dependence on fossil fuels and lowers carbon emissions
- Improves energy efficiency through tracking and MPPT, minimizing energy wastage
- Uses low-power electronic components to reduce overall energy consumption

The system supports the global shift toward green energy solutions and sustainable development.

5.2.3 Considerations to cultural and societal needs

The design also considers societal and practical aspects:

- Cost-effective implementation makes it suitable for developing countries like Bangladesh
- Can be used in rural or off-grid areas where electricity access is limited
- Simple design allows easy maintenance and understanding for educational purposes
- Encourages awareness and adoption of renewable energy technologies

By addressing affordability and usability, the system aligns with the needs of society and promotes wider acceptance of solar technology.

5.3 Investigations

The project investigated the implementation of an active mechatronic system designed to maintain a continuous 90° angle between a solar panel and the light source. Key areas of investigation included:

- **Differential Light Sensing:** Using a 4-quadrant Light Dependent Resistor (LDR) array to detect light intensity differentials across opposing vectors (North/South for elevation and East/West for azimuth).
- **Algorithmic Control:** Implementing logic that incrementally rotates high-torque servo motors toward the quadrant with the highest light intensity and utilizes a deadzone filter to prevent jitter and conserve power.
- **Power Optimization:** Integrating a Maximum Power Point Tracking (MPPT) boost converter to extract the maximum available power from the panel and step up the solar input to a regulated higher voltage.
- **Performance Gains:** The system was investigated for its efficiency, achieving a measured 13.6% relative performance gain over static solar installations.

5.3.1 Data Collection

Situation	Voltage Output	Panel Rotation	Temperature
Complete Dark	0V	No Rotation	Ok
Room Light	1.34V	No Rotation	Ok
Mobile Torch	5.36V	Rotation according to hand	Ok
Peak Sunlight	7.64V	Fixed at the sun direction	High

5.3.2 Results and Analysis

The comparative data between the **Static System** and the **Dual-Axis Tracker** highlights the following key findings:

- **Higher Voltage Yield:** The tracking system achieved an average output of **5.09V**, significantly outperforming the static panel's average of **4.48V**.
- **Peak Efficiency:** During maximum solar intensity, the tracker reached a peak of **5.23V**, representing **efficiency gain**.
- **Consistent Output:** While the static panel's performance fluctuated based on the sun's position, the dual-axis tracker maintained a stable, high-voltage curve by constantly optimizing the angle of incidence.
- **Operational Success:** The 4-quadrant LDR logic and 100 ADC dead zone successfully balanced precise tracking with power conservation, proving that active tracking effectively maximizes solar harvesting potential.

5.4 Limitations of Tools

While the system provides significant efficiency improvements, it faces several technical limitations:

- **Mechanical Range Constraints:** The servo motors are clamped to specific movement ranges, with the elevation axis limited to 10°–170° and the azimuth axis limited to 30°–160°.
- **Threshold Sensitivity:** The system relies on a dead zone filter (ignoring differences ≤ 100 ADC counts) and a minimum solar threshold (Solar Value > 200) to activate its tracking and cooling logic.
- **Component Specificity:** The current implementation is designed for a specific hardware architecture, including the Arduino UNO and SG90 servos, which may limit the weight or size of the solar panels it can effectively maneuver.
- **Environmental Reliability:** Although the system includes automated temperature control via a cooling fan, its overall long-term durability in extreme outdoor weather conditions remains a factor for further consideration.

5.5Impact Assessment

This section evaluates the professional responsibilities associated with the Smart Dual Axis Solar Tracking system beyond its technical specifications.

5.5.1 Assessment of Societal and Cultural Issues

The project promotes renewable energy adoption, making solar power more efficient and cost-effective for local residential and agricultural use. It fosters a cultural shift toward sustainable "smart" technology. Implementation must consider the aesthetic and spatial integration of moving mechanical structures within existing urban or rural environments.

5.5.2 Assessment of Health and Safety Issues

Potential safety risks include mechanical pinching from moving parts and electrical hazards. These are mitigated through software-defined rotation limits and automated temperature monitoring with active fan cooling. By providing a clean energy alternative, the system indirectly supports public health by reducing reliance on air-polluting fossil fuels.

5.5.3 Assessment of Legal Issues

The system complies with standard electrical safety codes and electronic waste guidelines. Deployment adheres to local building regulations for outdoor installations and utilizes open-source software licenses. Future scaling would require compliance with grid-connection laws and land-use regulations regarding structural height and neighborly "right to light."

5.6 Sustainability Evaluation

This section assesses the environmental and resource sustainability of the Smart Dual Axis Solar Tracking system.

- **Environmental Impact:** The system maximizes clean energy harvesting, achieving a 13.6% efficiency increase over static panels. This reduces the total number of panels and land area required for power generation, effectively lowering the overall carbon footprint and resource consumption.
- **Resource & Waste Management:** By utilizing an MPPT-based boost converter, the design ensures minimal energy wastage. The project uses modular electronic components, allowing for specific repairs or upgrades (such as replacing a servo or sensor) rather than disposing of the entire unit, which supports electronic waste reduction.
- **Long-Term Viability:** The use of low-power microcontrollers and automated tracking makes the system sustainable for remote, off-grid applications. It provides an economically viable solution by accelerating the return on investment through increased power yield and reduced manual maintenance requirements.

5.7 Ethical Issues

- **Integrity in Reporting:** All performance data, including the 13.6% efficiency gain, was recorded honestly. No technical results or sensor readings were fabricated to exaggerate the system’s success.
- **Public Safety & Welfare:** The design prioritizes safety through software movement limits and thermal management (automatic cooling). This aligns with the engineer's duty to protect users and the environment.
- **Professional Accountability:** Credit is accurately attributed to all group members based on their specific contributions to hardware, coding, and testing. All external libraries and resources are used in compliance with academic integrity standards.
- **Transparency:** Sourcing of components was done transparently, and the project is documented to be reproducible, supporting the ethical principle of knowledge sharing within the engineering community.

6 Reflection on Individual and Team work

6.1Individual Contribution of Each Member

Student ID	Name	Contribution
2106016	Mashrafi Al Asif	Frame design, Problem debugging, participated hardware setup and testing
2106019	MD Shahriar Sami	Frame design, Problem debugging, participated hardware setup and testing
2106020	Saumik Bachar	Initial idea, Circuit Implementation, Coding, debugging, Component Collection, Testing
2106021	Sinha Rahman Shourov	Initial idea, Circuit Implementation, Frame idea, Component Collection, Debugging, Assembly
2106027	MD Ismail Hossain	Implementation idea, Temperature sensor and light using idea, hardware setup and testing, debugging

6.2Log Book of Project Implementation

Date	Planned Work	Work Done	Remarks/Results
Jan 06	Initial Team Meeting	Defined objectives and finalized the dual-axis scope.	Project officially started.
Jan 15	Literature Review	Researched LDR sensor arrays and MPPT boost topologies.	Selected 4-quadrant LDR design.
Jan 28	Component Sourcing	Purchased Arduino, SG90 servos, and solar panel components.	All parts verified and functional.
Feb 10	Mechanical Fabrication	Built the dual-axis rotatable frame for the solar panel.	Stable movement on both axes.

Feb 22	Circuit Assembly	Integrated LDR sensors, servos, and the MPPT module.	Circuit breadboarded for testing.
Mar 05	Logic Coding	Developed Arduino code for light tracking and deadzone filters.	Threshold set to 100 ADC counts.
Mar 15	Hardware Integration	Mounted electronics onto the frame and calibrated servo limits.	Elevation: 10°-170°; Azimuth: 30°-160°.
Mar 25	System Testing	Conducted outdoor sunlight tests and data logging.	**efficiency gain** observed.
Mar 30	Optimization	Refined the cooling fan activation logic based on temperature.	Improved thermal stability.
Apr 03	Final Review	Prepared the final report and presentation slides.	**Project Completed.**

7 Communication to External Stakeholders

7.1 User Manual

Overview & Setup This system automatically aligns a solar panel with the sun using a 4-quadrant LDR sensor array and dual SG90 servos to maximize energy collection. To set up, place the unit on a flat, level surface with a clear view of the sky and connect a 9V–12V DC power source to the Arduino. Ensure the LDR sensor head is oriented in the same direction as the solar panel for accurate tracking.

Operation & Maintenance Once powered, the system scans for the brightest light source, adjusting position if the sensor differential exceeds 100 ADC counts. It includes an automated cooling fan for thermal management and software limits (Elevation: 10°–170°; Azimuth: 30°–160°) to protect the mechanical structure. Maintenance requires periodically cleaning the panel and sensors with a dry cloth and checking for secure wire connections to ensure smooth, unobstructed movement.

7.2 Github Link

GitHub - sinhashourov/Dual-Axis-Solar-Tracking-System-with-Maximum-Power-Point-Tracking-MPPT-: This project demonstrate automated dual axis solar tracker with maximum power point tracking. The code **and overall project report and youtube video link is provided. · GitHub**

7.3 YouTube Link

https://www.youtube.com/shorts/DAZyECpQqhY

7.4Project Management and Cost Analysis

Component Name	Quantity	Unit Price (BDT)	Total Price (BDT)
Arduino Uno	1	650	650
Solar Panel (5V)	1	250	250
SG90 Servo Motor	2	150	300
LDR (Light Dependent Resistor)	4	5	20
XL6009 Boost Converter	1	120	120
I2C LCD (16x2)	1	350	350
Cooling Fan (5V)	1	60	60
Li-ion Batteries (3.7V)	2	150	300
Battery Holder	1	40	40
Resistors, Wires, and PCB	--	150	150
Frame Materials (PVC/Wood)	--	200	200
Total Estimated Cost			**2,440 BDT**

8 Future Work

- **IoT Integration:** Adding a Wi-Fi module (like ESP32) for real-time remote data logging and mobile app monitoring.
- **Enhanced Algorithms:** Implementing advanced MPPT techniques like "Perturb and Observe" to further optimize power extraction in fluctuating weather.
- **Hybrid Tracking:** Combining LDR sensors with a Real-Time Clock (RTC) to allow the system to reset to East at night and maintain accuracy on cloudy days.
- **Structural Scaling:** Upgrading to high-torque stepper motors and weatherproof enclosures to support larger, industrial-grade solar panels for outdoor use.
- **Material Efficiency:** Using lightweight aluminum or carbon-fiber composites for the frame to reduce motor load and internal power consumption.

9 References

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